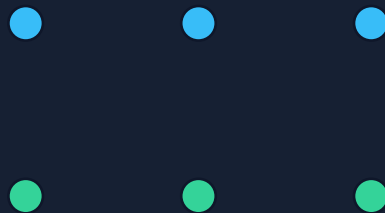


Data Cleaning Before vs After

How I manually cleaned a messy, inconsistent dataset into a clean, analysis-ready table using Microsoft Excel.

Swipe to see the results →



THE PROBLEM

The Raw, Messy Data

	A	B	C	D	E	F	G	H	I	J	K
	Customer ID	Customer Name	Phone Number	Email	Division	District	Area	Order Date	Sales Amount	Department	Employee Name
1	C1002	tanvir hossain	01711-326966	tanvir.hossain@outlook.com	rajshahi	Rajshahi	Paba	2025-03-16		IT	Sabbir
2	C1003	NUSRAT JAHAN		nusrat.jahan@outlook.com	DHAKA	Dhak	Mirpur	2025-03-24		IT	Mim Akter
3	C1004	Sadia Islam	01711-524011	sadia.islam@gmail.com	CHATTOGRAM	Chattogram	Agrabad	2025-04-30	16068	IT	ARAFAT HOSSAIN
4	C1005	Rahim Ahmed	01711733549	rahim.ahmed@gmail.com	Chattogram	Chattogram	Agrabad	31 Jan 2025		IT	Mim Akter
5	C1006	Tanvir Hossain	01711623473	tanvir.hossain@gmail.com	RAJSHAHI	Rajshahi	Paba	2025-05-24	24,334	IT	arafat hossain
6	C1007	rahim ahmed	01711-313110	rahim.ahmed@gmail.com	RAJSHAHI	Rajshahi	Paba	27 Feb 2025	19588	hr	Sabbir
7	C1008	Rahim Ahmed	01711327091	rahim.ahmed@yahoo.com	Dhaka	Narayanga	Fatullah	01/26/2025		HR	arafat hossain
8	C1009	Rahim Ahmed	1711219856	rahim.ahmed@yahoo.com	rajshahi	Rajshahi	Paba	30-04-2025	5441tk	hr	Mim Akter
9	C1010	Sabbir Alam	1711567563	sabbir.alam@outlook.com	CHATTOGRAM	Cumilla	Kotbari	02/05/2025		IT	ARAFAT HOSSAIN
10	C1011	Arif Khan	1711353495	arif.khan@gmail.com	Dhaka	Gazipur	Tongi	22 May 2025	13704	sales	Sabbir
11	C1012	Arif Khan	01711-738748		Chattogram	Chattogram	Agrabad	04/15/2025	45,466	IT	Arafat Hossain
12	C1013	tanvir hossain	01711451244	tanvir.hossain@gmail.com	dhaka	Narayanganj	Fatullah	2025-02-21		4989	Arafat Hossain
13	C1014	Jannat Mim		jannat.mim@gmail.com	CHATTOGRAM	Chattogram	Agrabad	29 Mar 2025		3370	sales
14	C1015	Arif Khan	1711104812	arif.khan@outlook.com	rajshahi	Rajshahi	Paba	2025-04-13	20,865	Sales	Arafat Hossain
15	C1016	Tanvir Hossain		tanvir.hossain@outlook.com	Chattogram	Cumilla	Kotbari	10-04-2025	1475	sales	arafat hossain
16	C1017	Rahim Ahmed	01711302782	rahim.ahmed@yahoo.com	chattogram	Chattogram	Agrabad	04/05/2025	18833tk	HR	ARAFAT HOSSAIN
17	C1018	Tanvir Hossain	01711 693789	tanvir.hossain@yahoo.com	DHAKA	Dhaka	Mirpur	09 Feb 2025	34785tk	HR	Sabbir
18	C1019	ARIF KHAN		arif.khan@gmail.com	Chattogram	Cumilla	Kotbari	07 Feb 2025		49450	IT
19	C1020	Karim Hasan	01711-247478	karim.hasan@outlook.com	Chattogram	Chattogram	Agrabad	2025-01-16	43964tk	sales	Arafat Hossain
20	C1021	sabbir alam		sabbir.alam@gmail.com	dhaka	Gazipur	Tongi	10 May 2025	47,756	Sales	Arafat Hossain
21	C1022	arif khan	01711-454133	arif.khan@gmail.com	Rajshahi	Rajshahi	Paba	04/09/2025	37600tk	IT	arafat hossain
22	C1023	TANVIR HOSSAIN	01711582254	tanvir.hossain@gmail.com	Chattogram	Chattogram	Agrabad	01/11/2025	37,627	HR	Arafat Hossain
23	C1024	Karim Hasan	01711-975773	karim.hasan@yahoo.com	RAJSHAHI	Bogura	Sherpur	2025-03-16	8902tk	sales	arafat hossain
24	C1025	TANVIR HOSSAIN	1711925288	tanvir.hossain@yahoo.com	Chattogram	Cumilla	Kotbari	02-01-2025		44027	Sales
25	C1026	Jannat Mim	01711249792	jannat.mim@gmail.com	Chattogram	Cumilla	Kotbari	04/07/2025		16586	sales
26	C1027	Nusrat Jahan	01711-945021	nusrat.jahan@outlook.com	RAJSHAHI	Bogura	Sherpur	2025-04-02	41782	IT	ARAFAT HOSSAIN
27	C1028	Karim Hasan	01711347055	karim.hasan@outlook.com	rajshahi	Bogura	Sherpur	2025-04-01	34456	IT	ARAFAT HOSSAIN
28	C1029	Sadia Islam	01711235628		rajshahi	Bogura	Sherpur	28 Jan 2025	16,719	HR	arafat hossain
29	C1030	Nusrat Jahan	01711 157404	nusrat.jahan@outlook.com	Dhaka	Narayanganj	Fatullah	2025-03-21		13373	IT
30	C1031	Sabbir Alam	01711 485873	sabbir.alam@gmail.com	Dhaka	Narayanganj	Fatullah	2025-02-16		43748	Marketing
31	C1032	Sabbir Alam		sabbir.alam@yahoo.com	rajshahi	Bogura	Sherpur	05/11/2025	47734tk	sales	Sabbir
32	C1033	Sabbir Alam	01711-311146	sabbir.alam@gmail.com	Rajshahi	Rajshahi	Paba	2025-05-08	7842tk	IT	Arafat Hossain
33	C1034	SABBIR ALAM	01711-541277	sabbir.alam@outlook.com	chattogram	Cumilla	Kotbari	10 Mar 2025	16,038	HR	arafat hossain
34	C1035	Sabbir Alam	01711 713230	sabbir.alam@gmail.com	Dhaka	Gazipur	Tongi	05/23/2025		IT	Sabbir
35	C1036	Sabbir Alam	01711-644511		Rajshahi	Rajshahi	Paba	2025-02-15	31,407	hr	Sabbir
36	C1037	karim hasan	01711-817444	karim.hasan@gmail.com	Dhaka	Gazipur	Tongi	2025-03-20	34323tk	HR	Sabbir
37	C1038	tanvir hossain	01711-482250	tanvir.hossain@gmail.com	DHAKA	Narayanganj	Fatullah	02-04-2025	11,147	hr	Mim Akter
38	C1039	sabbir alam	01711-654381	sabbir.alam@yahoo.com	Dhaka	Gazipur	Tongi	2025-05-28	39638tk	sales	Arafat Hossain
39	C1040	SABBIR ALAM	1711837131	sabbir.alam@gmail.com	Chattogram	Cumilla	Kotbari	02-02-2025	27,186	HR	Mim Akter
40	C1041	rahim ahmed	01711-422368		Rajshahi	Bogura	Sherpur	2025-05-23		12440	IT
41	C1042	tanvir hossain	1711853362	tanvir.hossain@gmail.com	dhaka	Gazipur	Tongi	18-05-2025		HR	ARAFAT HOSSAIN
42	C1043	RAHIM AHMED	01711-293054	rahim.ahmed@gmail.com	Dhaka	Narayanganj	Fatullah	2025-02-10	2923tk	IT	arafat hossain
43	C1044	arif khan	01711-305565	arif.khan@gmail.com	Chattogram	Chattogram	Agrabad	12-05-2025		HR	Arafat Hossain
44	C1045	Tanvir Hossain	01711761457	tanvir.hossain@gmail.com	Rajshahi	Rajshahi	Paba	03 Mar 2025	43,035	IT	Arafat Hossain
45	C1046	Nusrat Jahan	01711954080		Chattogram	Chattogram	Agrabad	2025-02-05	1855tk	hr	ARAFAT HOSSAIN

⚠ BEFORE - Raw, unstructured data with multiple issues

8 Data Quality Problems Identified

1

Inconsistent Customer IDs

Mixed uppercase/lowercase: C1001 vs c1001

2

Inconsistent Name Casing

RAHIM AHMED vs rahim ahmed vs Rahim Ahmed

3

Non-standard Date Formats

2025-05-16, 31 Jan 2025, 04/15/2025, 30-04-2025

4

Mixed Sales Amount Formats

Numbers stored as text: 47734tk, 2923tk, 5441tk

5

Missing Phone Numbers

Multiple blank cells in Phone Number column

6

Missing & Incorrect Email Addresses

Several rows with invalid & no email data

7

Inconsistent Division Names

RAJSHAHI vs Rajshahi vs rajshahi

8

Highlighted/Colored Rows

Random cell highlights - no clear logic or meaning

THE RESULT

The Clean, Analysis-Ready Data

	A	B	C	D	E	F	G	H	I	J	K
1	Customer ID	Customer Name	Phone Number	Email	Division	District	Area	Order Date	Sales Amount	Department	Employee Name
2	C1024	Karim Hasan	01711975773	karim.hasan@yahoo.com	Rajshahi	Bogura	Sherpur	16/Mar/25	\$ 8,902	Sales	Arafat Hossain
3	C1027	Nusrat Jahan	01711945021	nusrat.jahan@outlook.com	Rajshahi	Bogura	Sherpur	02/Apr/25	\$ 41,782	IT	Arafat Hossain
4	C1028	Karim Hasan	01711347055	karim.hasan@outlook.com	Rajshahi	Bogura	Sherpur	01/Apr/25	\$ 34,456	IT	Arafat Hossain
5	C1029	Sadia Islam	01711235628	Not Found	Rajshahi	Bogura	Sherpur	28/Jun/25	\$ 16,719	HR	Arafat Hossain
6	C1032	Sabbir Alam	Not Found	sabbir.alam@yahoo.com	Rajshahi	Bogura	Sherpur	05/Nov/25	\$ 47,734	Sales	Sabbir
7	C1041	Rahim Ahmed	01711422368	Not Found	Rajshahi	Bogura	Sherpur	23/May/25	\$ 12,440	HR	Arafat Hossain
8	C1050	Nusrat Jahan	01711936731	Not Found	Rajshahi	Bogura	Sherpur	17/Apr/25	\$ 32,177	IT	Mim Akter
9	C1053	Nusrat Jahan	01711456583	nusrat.jahan@gmail.com	Rajshahi	Bogura	Sherpur	02/Mar/25	\$ 15,662	Marketing	Arafat Hossain
10	C1054	Jannat Mim	Not Found	jannat.mim@gmail.com	Rajshahi	Bogura	Sherpur	02/Apr/25	\$ 33,983	Marketing	Arafat Hossain
11	C1056	Rahim Ahmed	01711 840961	rahim.ahmed@yahoo.com	Rajshahi	Bogura	Sherpur	03/May/25	\$ -	Sales	Mim Akter
12	C1065	Nusrat Jahan	01711 804290	nusrat.jahan@yahoo.com	Rajshahi	Bogura	Sherpur	29/Apr/25	\$ 2,580	Marketing	Sabbir
13	C1069	Karim Hasan	01711403026	karim.hasan@yahoo.com	Rajshahi	Bogura	Sherpur	26/Apr/25	\$ 8,927	Sales	Mim Akter
14	C1089	Rahim Ahmed	01711350964	Not Found	Rajshahi	Bogura	Sherpur	31/Jun/25	\$ 48,025	HR	Arafat Hossain
15	C1091	Sabbir Alam	Not Found	sabbir.alam@outlook.com	Rajshahi	Bogura	Sherpur	23/Mar/25	\$ 12,130	Marketing	Arafat Hossain
16	C1099	Rahim Ahmed	Not Found	rahim.ahmed@outlook.com	Rajshahi	Bogura	Sherpur	13/Jun/25	\$ 10,686	IT	Sabbir
17	C1004	Sadia Islam	01711524011	sadia.islam@gmail.com	Chattogram	Chattogram	Agrabad	30/Apr/25	\$ 16,068	IT	Arafat Hossain
18	C1005	Rahim Ahmed	01711733549	rahim.ahmed@gmail.com	Chattogram	Chattogram	Agrabad	31/Jun/25	\$ 30,684	IT	Mim Akter
19	C1012	Arif Khan	01711738748	Not Found	Chattogram	Chattogram	Agrabad	15/Apr/25	\$ 45,466	IT	Arafat Hossain
20	C1014	Jannat Mim	Not Found	jannat.mim@gmail.com	Chattogram	Chattogram	Agrabad	29/Mar/25	\$ 3,370	Sales	Sabbir
21	C1017	Rahim Ahmed	01711302782	rahim.ahmed@yahoo.com	Chattogram	Chattogram	Agrabad	04/May/25	\$ 18,833	HR	Arafat Hossain
22	C1020	Karim Hasan	01711247478	karim.hasan@outlook.com	Chattogram	Chattogram	Agrabad	16/Jun/25	\$ 43,964	Sales	Arafat Hossain
23	C1023	Tanvir Hossain	01711582254	Not Found	Chattogram	Chattogram	Agrabad	01/Nov/25	\$ 37,627	HR	Arafat Hossain
24	C1044	Arif Khan	01711305565	arif.khan@gmail.com	Chattogram	Chattogram	Agrabad	12/May/25	\$ -	IT	Arafat Hossain
25	C1046	Nusrat Jahan	01711954080	Not Found	Chattogram	Chattogram	Agrabad	05/Feb/25	\$ 1,855	HR	Arafat Hossain
26	C1060	Jannat Mim	01711 311764	Not Found	Chattogram	Chattogram	Agrabad	05/Apr/25	\$ 27,762	Sales	Arafat Hossain
27	C1061	Karim Hasan	01711 698109	karim.hasan@outlook.com	Chattogram	Chattogram	Agrabad	06/Jun/25	\$ 31,215	Marketing	Arafat Hossain
28	C1067	Tanvir Hossain	Not Found	tanvir.hossain@gmail.com	Chattogram	Chattogram	Agrabad	01/Apr/25	\$ 28,168	HR	Mim Akter
29	C1082	Rahim Ahmed	Not Found	rahim.ahmed@yahoo.com	Chattogram	Chattogram	Agrabad	24/Apr/25	\$ 35,760	HR	Sabbir
30	C1085	Arif Khan	01711541071	arif.khan@yahoo.com	Chattogram	Chattogram	Agrabad	07/Mar/25	\$ -	Marketing	Sabbir
31	C1090	Arif Khan	01711981066	arif.khan@yahoo.com	Chattogram	Chattogram	Agrabad	01/Sep/25	\$ -	HR	Arafat Hossain
32	C1093	Tanvir Hossain	Not Found	tanvir.hossain@gmail.com	Chattogram	Chattogram	Agrabad	23/Sep/25	\$ -	IT	Sabbir

✓ AFTER - Consistent, clean, fully structured and ready for analysis

WHAT WAS FIXED

Every Fix, Step by Step



Standardized Customer IDs

All IDs now follow consistent C#### uppercase format



Normalized Name Casing

PROPER CASE applied - Rahim Ahmed, Jannat Mim



Unified Date Format

All dates converted to DD/Mon/YY - e.g. 16/Mar/25



Cleaned Sales Amounts

Removed 'tk' suffix, converted to numeric with \$ sign



Handled Missing Phones

Marked as “Not Found” - clearly flagged for follow-up



Handled Missing Emails

Missing values set as “Not Found” and invalid entries corrected



Standardized Division Names

Proper Case: Rajshahi, Chattogram, Dhaka throughout



Removed All Cell Highlights

Clean white background - no misleading color coding